

Design Patterns Final Exam Cheat Sheet

Important MCQs

1. Template Method pattern defines:
 - A. Skeleton of algorithm
2. Hooks in Template Method allow:
 - C. Optional behavior
3. Strategy pattern is used to:
 - C. Encapsulate interchangeable behavior
4. Adapter pattern:
 - B. Converts interface of one class into another
5. Facade pattern provides:
 - B. Simplified interface to complex subsystem
6. MVC's Controller:
 - C. Handles input and updates Model/View
7. In MVC, data logic is handled by:
 - B. Model

Important Conceptual Questions

- Explain the Template Method Pattern with example.
- How does the Strategy Pattern support open/closed principle?
- Compare Adapter and Facade patterns with real-world examples.
- Why is MVC considered an architectural pattern?
- Explain the flow of control in an MVC application.

Coding Practice Problems

- Implement a Template Method pattern for an OnlineTransaction system (Bank, Credit, Crypto).
- Write Java code for adapting a Turkey class to behave like a Duck.

- Implement Strategy Pattern for different fee calculations (Standard, Premium, Crypto).
- Create a Facade class for a Home Theater system that includes DVD, Projector, Lights.
- Code a simple MVC example for a Bookstore (search, view, checkout).

Key Concepts Summary

- Template Method: Skeleton in superclass, variations in subclasses. Allows hooks.
- Strategy: Encapsulates behaviors, allows runtime changes. Composition over inheritance.
- Adapter: Converts incompatible interfaces. Used to wrap legacy or third-party code.
- Facade: Simplifies interaction with complex subsystems. Provides default setup.
- MVC: Model = data logic, View = UI, Controller = input logic. Promotes separation of concerns.